

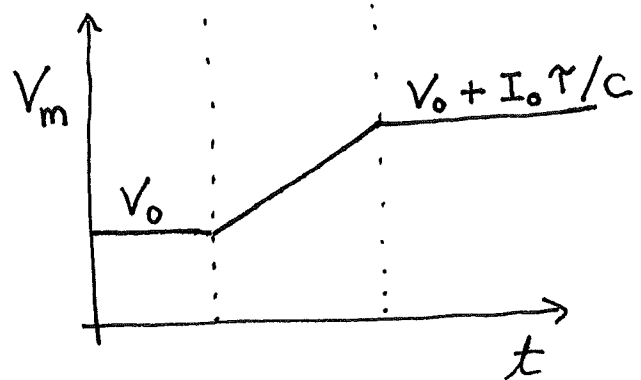
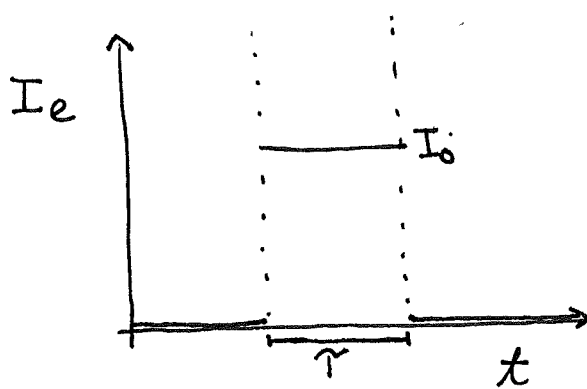
Current integration in "capacitor model" of neuron.

$$C \frac{dV_m}{dt} = I_e$$

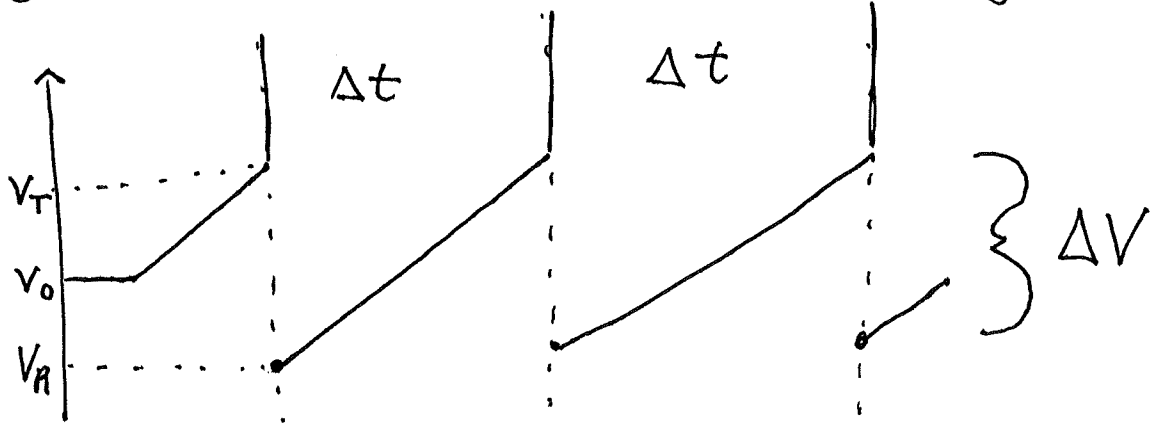
$$V_m(t) = V_m(0) + \frac{1}{C} \int_0^t dt' I_e(t')$$

$$\Delta V_m = \frac{\Delta Q}{C}$$

Response to a current step:



A simple model of neuronal firing: Integrate & Fire



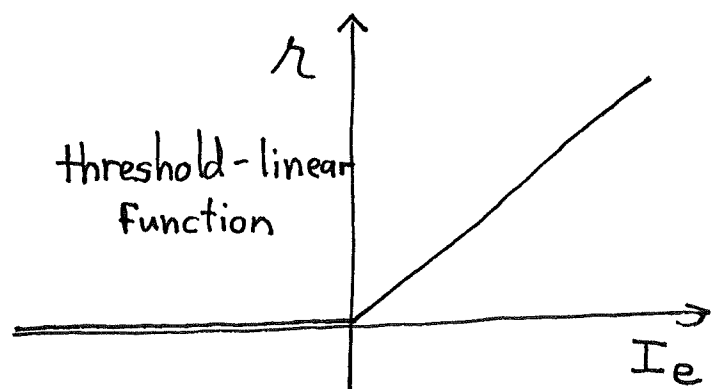
Firing rate:

$$h = \frac{1}{\Delta t}$$

$$\Delta V = V_T - V_R = \frac{I_e \Delta t}{C}$$

$$h = \frac{I_e}{C(V_T - V_R)}$$

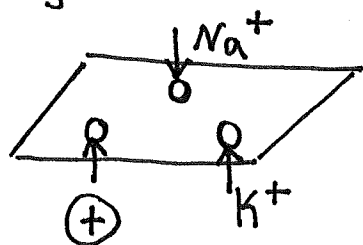
F-I curve



Resistance $\Rightarrow$ Leaky integrate & fire

$$I_e - I_c - I_L = 0$$

Why isn't resistance infinite? Holes let charge through. This is termed leak.

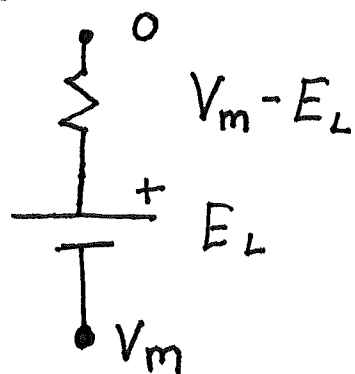


Different holes
Different charged ions $\Rightarrow E_L \neq 0$
(Battery)

Ohm's law:

$$I_L = \frac{V}{R_L}$$

Resistor & battery



$$V_m = I_L R_L + E_L$$

$$I_L = \frac{V_m - E_L}{R_L} = G_L (V_m - E_L)$$

"Leaky integrator model" of neuron:

$$C \frac{dV_m}{dt} = I_e - G_L (V_m - E_L)$$

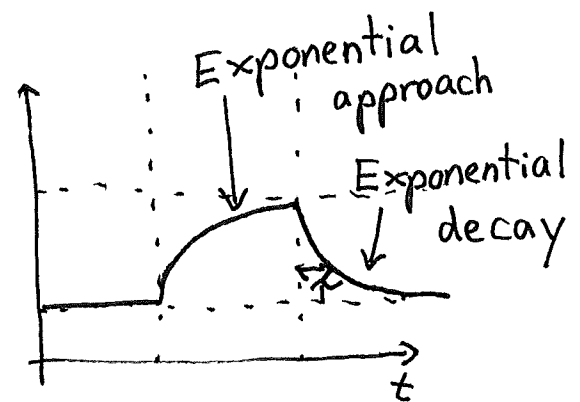
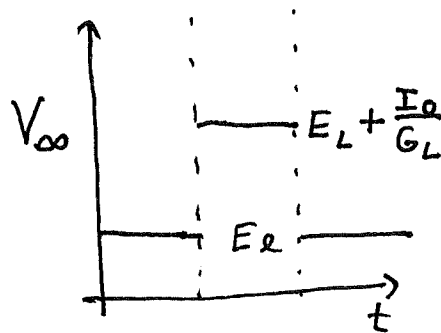
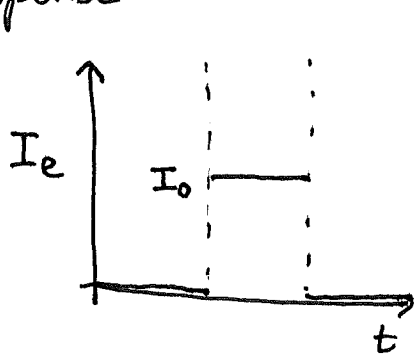
Current integration in "leaky integrator" model of a neuron.

$$C \frac{dV_m}{dt} = I_e - G_L (V_m - E_L)$$

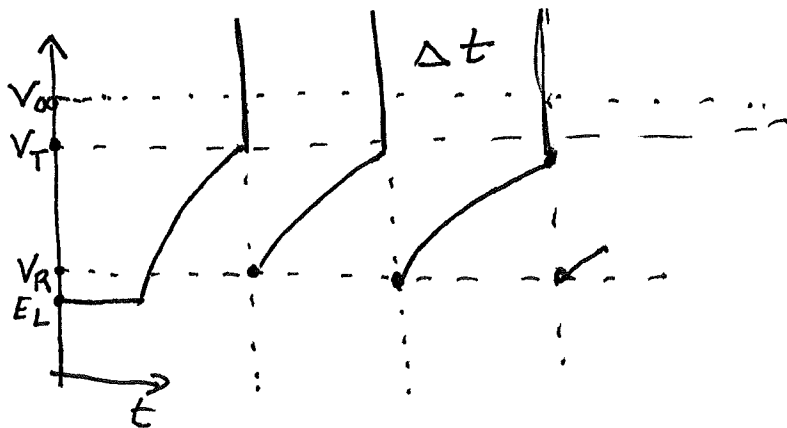
$$\underbrace{\frac{C}{G_L} \frac{dV_m}{dt}}_{\text{RC time constant}} = -V_m + \underbrace{\frac{I_e}{G_L} + E_L}_{\text{Steady-state voltage}}$$

$$\tau \frac{dV_m}{dt} = -V_m + V_\infty$$

Response to current step:



Back to our simple model of spiking



Threshold current for spiking:

$$V_T = V_\infty = \frac{I_T}{G_L} + E_L$$

$$I_T = G_L (V_T - E_L)$$

Above threshold:

$$V_m(0) = V_R, \quad V_m(\Delta t) = V_T$$

$$V_m(t) = V_m(0)e^{-t/\tau} + (1 - e^{-t/\tau})V_\infty$$

Solve for Δt :

$$V_T = V_R e^{-\Delta t/\tau} + (1 - e^{-\Delta t/\tau})V_\infty$$

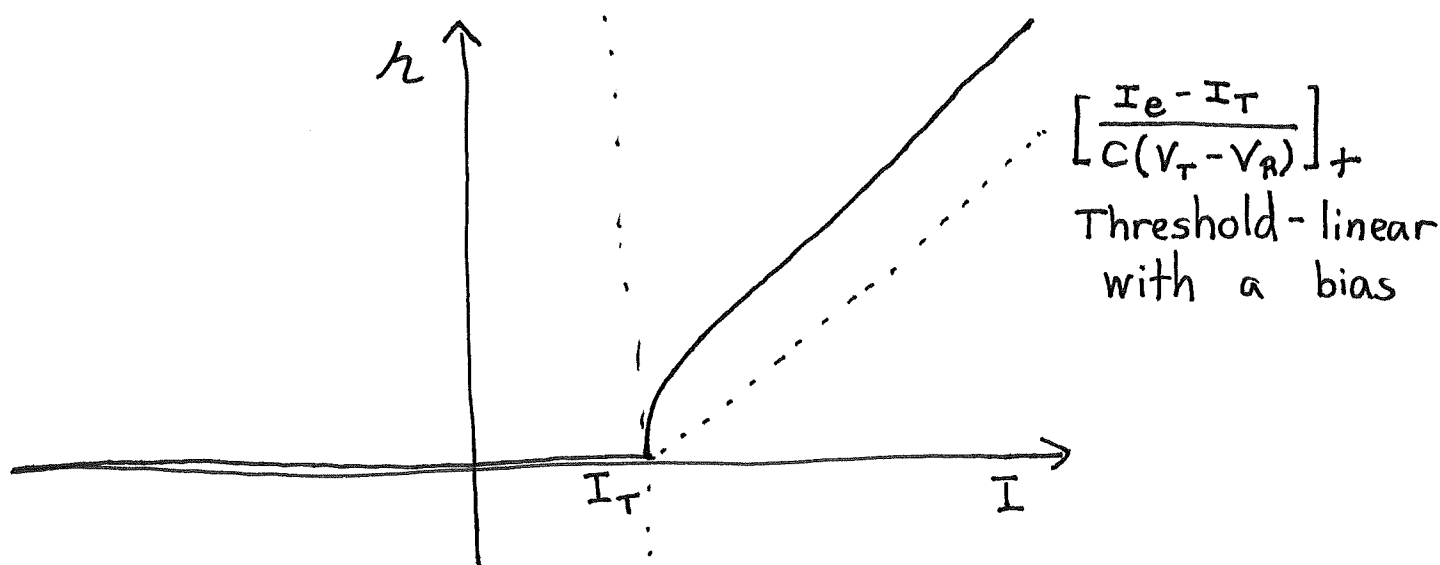
$$V_T - V_\infty = (V_R - V_\infty)e^{-\Delta t/\tau} \Rightarrow e^{-\Delta t/\tau} = \frac{V_T - V_\infty}{V_R - V_\infty}$$

$$\Rightarrow \Delta t = -\tau \ln \frac{V_T - V_\infty}{V_R - V_\infty} = \tau \ln \frac{V_R - V_\infty}{V_T - V_\infty}$$

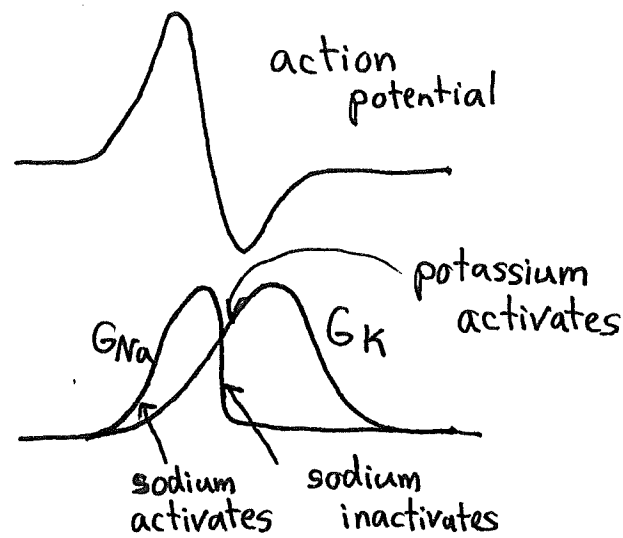
F-I curve:

$$h = \frac{1}{\Delta t} = \frac{1}{\tau \ln[(V_\infty - V_R)/(V_\infty - V_T)]}$$

$$= \frac{G_L}{C \ln[(I_e/G_L + E_L - V_R)/(I_e/G_L + E_L - V_T)]}$$



Sodium & Potassium conductances during action potential



Hodgkin-Huxley model of action potential:

$$C \frac{dV_m}{dt} = I_e - G_L (V_m - E_L) - \bar{G}_{Na} m^3 h (V_m - E_{Na}) - \bar{G}_K (V_m - E_K) n^4$$

m : Na activation; h : sodium inactivation; n : potassium activation

$\bar{G}_{Na}$, $\bar{G}_K$: maximal conductances

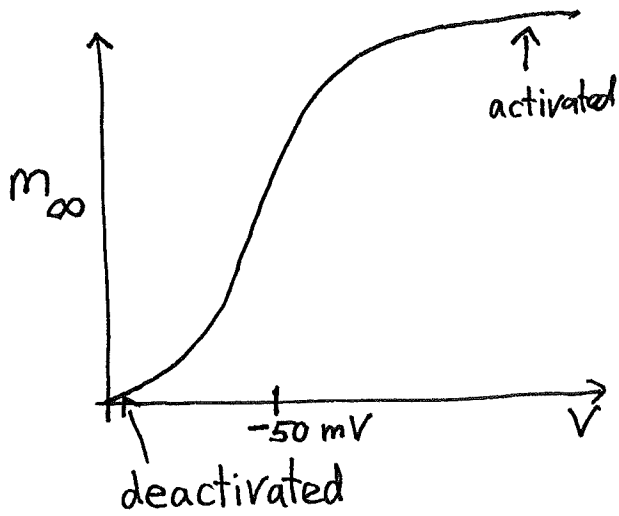
Time-dependence of gating variables:

$$\begin{aligned} \frac{dn}{dt} &= \alpha_n (1 - n) - \beta_n n \\ &= -(\alpha_n + \beta_n) n + \alpha_n \end{aligned}$$

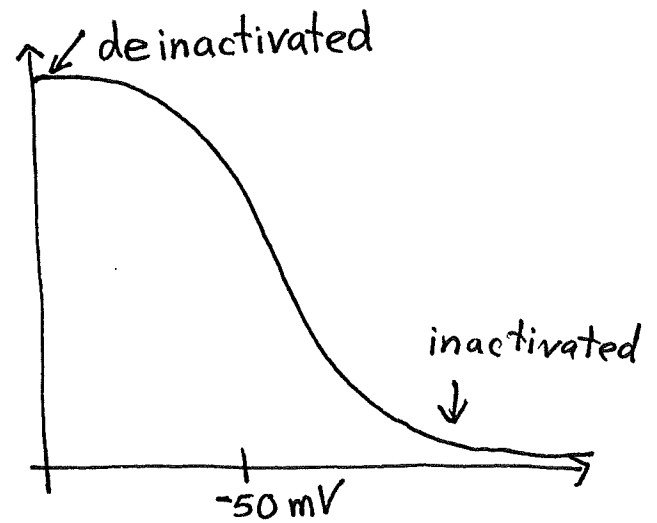
$$\tau_n \frac{dn}{dt} = -n + n_\infty ; \quad \tau_n = \frac{1}{\alpha_n + \beta_n} \quad n_\infty = \frac{\alpha_n}{\alpha_n + \beta_n}$$

Note that α_n , β_n , τ_n , n_∞ are all voltage dependent! Requires simulation.

Sodium "activates" at high V



Sodium "inactivates" at high V



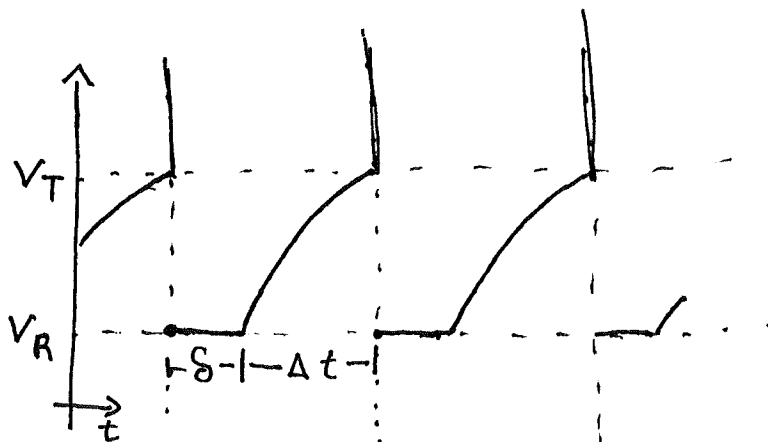
Potassium similarly activates at high V , but it does not inactivate.

Refractory period:

Sodium de-inactivation takes time, so action potentials can't be too closely spaced. F-I curves thus saturate at high currents.

Refractory periods integrate-and-fire models:

$$\mu = \frac{1}{\delta + \Delta t} \quad ; \quad \delta: \text{refractory period} \\ \Delta t: \text{derived earlier}$$



$$\lim_{I_0 \rightarrow \infty} \mu = \frac{1}{\delta}$$

Important remarks:

- Membrane capacitance & conductances scale with membrane area. It's convenient to normalize.

$$C = \frac{C}{A}, g_L = \frac{G_L}{A}, \text{ etc.}$$

- Can account for neuron morphology: cable equation; multicompartment modeling
- Primacy of action potentials: subthreshold signals attenuate, action potentials propagate
- Cellular diversity: different cells have different channels and morphologies. The Hodgkin-Huxley framework carries over.